Greedy Algorithm

$n = 7 \rightarrow$ no of objects

$m = 15 \rightarrow$ Bag capacity

Knapsack problem

Object O:	1	2	3	4	5	6	7
Profit P:	10	5	15	7	6	18	3
weight w:	2	3	5	7	1	4	1

Profit should be maximum
weight should be under 15

Based on max profit

objects	Profit (P)	weight (w)	Remaining weight
3	15	5	15 - 5 = 10
2	10	3	10 - 3 = 7
6	9	3	7 - 3 = 4
5	8	1	4 - 1 = 3
4	$7 \times \frac{3}{4} = 5.25$	3	3 - 3 = 0
	47.25		

maximum weised

Based on ~~min~~ weight

46

object	Profit (P)	weight (w)	Remaining weight
1	5	1	15 - 1 = 14
5	8	1	14 - 1 = 13
7	4	2	13 - 2 = 11
2	10	3	11 - 3 = 8
6	9	3	8 - 3 = 5
4	7	4	5 - 4 = 1
3	$15 \times \frac{1}{5} = 3$	1	1 - 1 = 0

maximum Profit/weight Ratio

object :-	1	2	3	4	5	6	7
Profit (P) :-	5	10	15	7	8	9	4
weight (w) :-	1	3	5	4	1	3	2
$\frac{\text{Profit}}{\text{weight}}$:-	5	3.3	3	1.75	8	3	2

objects	Profit (P)	weight (w)	Remaining weight
5	8	1	15-1 = 14
1	5	1	14-1 = 13
2	10	3	13-3 = 10
3	15	5	10-5 = 5
6	9	3	5-3 = 2
7	4	2	2-2 = 0
	51		

maximum profit ⟶ 47.25

menimum weight ⟶ 46

maximum $\frac{\text{Profit}}{\text{weight}}$ ratio ⟶ 51 ✓